SREYAS MOHAN

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EDUCATION

- **New York University** — New York City, USA
  PhD candidate at the **Center for Data Science**; **CGPA**: 4.0/4.0 — Sept 2017 - Present
  - Advised by Prof. Carlos Fernandez-Granda and Prof. Eero P Simoncelli
  - Key Courses: *Mathematics of Deep Learning, Deep Learning, Inference and Representation, Optimization Based Data Analysis, Natural Language Processing With Representation Learning*
- **Indian Institute of Technology, Madras** — Chennai, India
  B.Tech in Electrical Engineering; **CGPA**: 9.02/10 — Aug 2013 – Jul 2017
  - Key Courses: *Machine Learning, Reinforcement Learning, Machine Learning for Computer Vision, Probability, Statistics and Stochastic Processes, Applied Linear Algebra for EE, Digital Signal Processing*

PUBLICATIONS & PREPRINTS

- "Deep Unsupervised Video Denoising." [ArXiv]
  D. Sheth, **S. Mohan***, *J. Vincent, R. Monzorro, P. Crozier, M. Khapra, E. Simoncelli, C. Fernandez-Granda*. Preprint (2020)
- "Deep Denoising for Scientific Discovery: A Case Study in Electron Microscopy." [ArXiv]
  ***Sreyas Mohan**, R. Monzorro, J. Vincent, B. Tang, D. Sheth, E. Simoncelli, D. Matteson, P. Crozier, C. Fernandez-Granda*. Preprint (2020)
- "Be Like Water: Robustness to Extraneous Variables Via Adaptive Feature Normalization." [ArXiv]
  Aakash Kaku, **Sreyas Mohan***, *Avinash Parnandi, Heidi Schambra, Carlos Fernandez-Granda*. Preprint (2020)
- "Robust and Interpretable Blind Image Denoising via Bias-free Convolutional Neural Networks" [ArXiv]
  Sreyas Mohan, *Zahra Kadkhodaie*, Eero Simoncelli and Carlos Fernandez-Granda*. Proc. International Conference on Learning Representations (ICLR), 2020
- "Data-driven Estimation of Sinusoid Frequencies" [ArXiv]
  *Gautier Izacard, **Sreyas Mohan**, Carlos Fernandez-Granda*. Proc. 33rd Conference on Neural Information Processing Systems (NeurIPS), Vancouver (Canada) 2019.
- "Automatic Knee Segmentation using Diffusion Weighted MRI" [ArXiv]
  A Duarte, C Hegde*, A Kaku*, **S Mohan***, *J G. Raya* Medical Imaging Meets NeurIPS, Neural Information Processing Systems (NeurIPS), Vancouver (Canada) 2019.
- "Blind Nonnegative Source Separation Using Biological Neural Networks" [ArXiv]
  *Cengiz Pehlevan, **Sreyas Mohan**, Dmitri Chklovskii.* Neural computation, vol. 29, no. 11, pp. 29252954, 2017. [EXTENDED ABSTRACT] presented at Cosyne 2017, Salt Lake City, USA.
- "Data Driven Coded Aperture Design for Depth Recovery" [ArXiv]
  *Prasan Shedligeri, **Sreyas Mohan**, Kaushik Mitra.* ICIP, pp. 56-60, 2017

∗ *equal contribution*

SCHOLASTIC ACHIEVEMENTS

- Awarded *Subramanian Rajalakshmi Indira (SRI)* Prize for the **Best Interdisciplinary Project** across all students in the graduating batch of 2017 at IIT Madras.
- Awarded DAAD[1] WISE Scholarship by **Goverment of Germany** in 2016. (Declined)

[1] DAAD is the German Academic Exchange Service. DAAD WISE

- Awarded **KVPY**(Kishore Vaigyanik Protsahan Yojana)[2] scholarship (2011) funded by the Department of Science and Technology, Government of India. (All India Rank 166)
- Travel Grant: NeurIPS (2019, 2020) Cosyne 2017,
- One among **70 students** selected **nationwide** for NIUS [3] Fellowship in 2013.

PROFESSIONAL EXPERIENCE

Visiting Researcher
- *Institute of Science and Technology, Austria* May – July 2017
  Dr. Gasper Tkacik
  - Deep learning model to reconstruct a complex movie from the activities of the ganglion cells in a mammalian retina.

Strategy Researcher
- *Trexquant Investment LP, Stamford, USA* Jan – April 2017
  Worked remotely from India
  - Designed and implemented machine-learning algorithms to filter and assign weights to thousands of proprietary return forecasts of stocks in Trexquant's database.

Research Intern
- *Simons Center for Data Analysis, New York* May – Aug 2016
  Dr. Dmitri Chklovskii, Dr. Cengiz Pehlevan
  - Worked on a biologically plausible two-layer neural network that learns to separate mixtures of non-negative sources to its components. In departure from previous approaches, our network is derived from a similarity matching principle.

Data Science Intern
- *Lighthouse Datalab, Mumbai, India* May – Jul 2015
  Targeted Advertising. Mentored by Dr. Sriram Subramnian. [REPORT]

Computer Vision Intern
- *HyperVerge, Chennai, India* May – Jul 2014
  Detecting Fire and Smoke from low-resolution CCTV Camera Feeds.

Android App Developer
- *Clozerr, Chennai, India* Feb – Sep 2014
  Android and Web Application Developer. Led a team of 5 people.

TEACHING EXPERIENCE

At New York University:
- *DS-GA 1013 Mathematical Tools for Data Science* (Spring 2020, 2021) for Prof. Carlos Fernandez-Granda
- *DS-GA 1002 Probability and Statistics* (Fall 2020) for Prof. Carlos Fernandez-Granda and Prof. Brett Bernstein
- *DS-GA 1011 Natural Language Processing with Representation Learning* (Fall 2019) for Prof. Kyunghyun Cho
- *DS-GA 1003 Machine Learning* (Spring 2019) for Prof. Julia Kempe and Dr. David Rosenberg.
- *CSCI-GA.1170 Fundamental Algorithms* (Summer, Fall 2018) for Prof. Alexander Alekseyev

Elsewhere:
- **Kigali, Rwanda**: NLP with Deep Learning at the African Institute of Mathematical Sciences, Rwanda. (March 2019, 2020). For Prof. Kyunghyun Cho
- **IIT Madras, India**: *EE5177 Machine Learning for Computer Vision* (Spring 2017) for Prof. Kaushik Mitra
- **Chennai, India**: IViL[4] - Volunteered to teach Physics and Mathematics for Higher Secondary School students from underprivileged sections of the society (2013-2017)

SERVICE

Reviwer for ICLR (2020, 2021), NeurIPS (2021) and, Journal of Electronic Imaging (JEI)

[2] Translated as Young Scientist Encouragement Program. KVPY
[3] NIUS is instituted by HBCSE(Tata Institute for Fundamental Research) Bombay
[4] IIT for Villages-student run organisation with the vision of helping rural India develop by leveraging the resource base at IITM